

Version 2 - XGBoost Safe Plus Rolling HIGH_ACCURACY - Feature Used / Dropped Audit

เทียบ source clean dataset กับ df_engineered และระบุว่า feature ใดใช้ train model / feature ใดตัดทิ้ง

Model Summary

Model	Dataset	df_engineered	Used Features	Accuracy	Recall	F1	AUC	Rating
XGBoost safe rolling	clean_dataset_v2_high_signal.csv	df_engineered_v2_HIGH_ACCURACY.csv	71	88.88%	76.03%	79.99%	94.66%	A

Source Artifacts

artifact	path	columns	sample_rows_read
source_data	docs\version 2\2_xgboost_safe_plus_rolling_HIGH_ACCURACY\data\clean_dataset_v2_high_signal.csv	65	5
df_engineered	docs\version 2\2_xgboost_safe_plus_rolling_HIGH_ACCURACY\data\df_engineered_v2_HIGH_ACCURACY.csv	76	5
used_feature_list	docs\version 2\2_xgboost_safe_plus_rolling_HIGH_ACCURACY\data\used_features_v2_xgboost_safe_plus_rolling_HIGH_ACCURACY.csv	1	5

Used Feature Groups

group	feature_count	example_features
customer_history	22	hist_order_count, hist_order_count_180d, hist_order_count_30d, hist_order_count_365d, hist_order_count_60d
price_discount	11	amount_per_item, campaign_discount_pct, discount_amount_ratio, discount_applied_amount, is_high_discount
time	10	avg_delivery_days, customer_age_days, customer_tenure_months, days_since_last_order, days_since_last_return
other	9	courier_type, high_value_order, is_bank_transfer, is_cod, is_credit_card
product	7	brand, category, category_channel, category_payment, is_fragile
customer_profile	5	age, age_group, damage_rate, gender, membership_tier
order_channel_payment	5	channel_type, payment_method, preferred_channel, province_payment, tier_payment
geography	2	coverage_region, province

Used Features (sample)

age, age_group, amount_per_item, avg_delivery_days, brand, campaign_discount_pct, category, category_channel, category_payment, channel_type, courier_type, coverage_region, customer_age_days, customer_tenure_months, damage_rate, days_since_last_order, days_since_last_return, delivery_time_expected_days, discount_amount_ratio, discount_applied_amount, gender, high_value_order, hist_order_count, hist_order_count_180d, hist_order_count_30d, hist_order_count_365d, hist_order_count_60d, hist_order_count_90d, hist_return_count_180d, hist_return_count_30d, hist_return_count_365d, hist_return_count_60d, hist_return_count_90d, hist_return_rate, hist_return_rate_180d, hist_return_rate_30d, hist_return_rate_365d, hist_return_rate_60d, hist_return_rate_90d, hist_spend_sum_180d, hist_spend_sum_30d, hist_spend_sum_365d, hist_spend_sum_60d, hist_spend_sum_90d, is_bank_transfer, is_cod, is_credit_card, is_fragile, is_high_discount, is_repurchased_item, is_weekend, log_total_amount, logistics_risk, low_rating_alert, membership_tier, order_dayofweek, order_hour, order_month, payment_method, preferred_channel ...

Dropped / Not Used Feature Logic

- คัด leakage/post-event/target fields จำนวนประมาณ 16 feature เช่น delay_days, delivery_date, delivery_days, expected_delivery_date, is_returned, item_condition, refund_amount, return_date, return_id, return_reason
- คัด identifier/high-cardinality fields จำนวนประมาณ 13 feature เช่น courier_id, courier_name, customer_id, customer_name, customer_phone, order_id, product_id, product_name, promo_id, promo_name

- feature ที่เป็น raw date หรือ raw text จะถูกแปลงเป็น numeric/encoded/rolling feature ก่อน หรือไม่ใช่ถ้าไม่ช่วย performance
- feature ที่เพิ่มแล้ว performance ไม่ดีขึ้นหรือซ้ำซ้อนกับ feature สำคัญ จะถูกจัดเป็น NOT USED เพื่อลด resource/query cost

Feature Audit Detail (sample rows)

Feature	In Clean	In Engineered	Used	Status	Reason
age	True	True	True	USED	model input feature
age_group	False	True	True	USED	model input feature
amount_per_item	False	True	True	USED	model input feature
avg_delivery_days	True	True	True	USED	model input feature
brand	True	True	True	USED	model input feature
campaign_discount_pct	True	True	True	USED	model input feature
category	True	True	True	USED	model input feature
category_channel	False	True	True	USED	model input feature
category_payment	False	True	True	USED	model input feature
channel_type	True	True	True	USED	model input feature
courier_type	True	True	True	USED	model input feature
coverage_region	True	True	True	USED	model input feature
courier_id	True	False	False	DROPPED	identifier or high-cardinality identity field
courier_name	True	False	False	DROPPED	identifier or high-cardinality identity field
customer_id	True	True	False	DROPPED	identifier or high-cardinality identity field
customer_name	True	False	False	DROPPED	identifier or high-cardinality identity field
customer_phone	True	False	False	DROPPED	identifier or high-cardinality identity field
dataset_split	False	True	False	NOT USED	engineered candidate not selected for this model
delay_days	True	False	False	DROPPED	post-event/target/leakage field
delivery_date	True	False	False	DROPPED	post-event/target/leakage field
delivery_days	True	False	False	DROPPED	post-event/target/leakage field
expected_delivery_date	True	False	False	DROPPED	post-event/target/leakage field
is_returned	True	True	False	DROPPED	post-event/target/leakage field
item_condition	True	False	False	DROPPED	post-event/target/leakage field
order_date	True	True	False	NOT USED	not selected in final feature set
order_id	True	True	False	DROPPED	identifier or high-cardinality identity field
product_id	True	False	False	DROPPED	identifier or high-cardinality identity field
product_name	True	False	False	DROPPED	identifier or high-cardinality identity field
promo_end_date	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
promo_id	True	False	False	DROPPED	identifier or high-cardinality identity field

Full CSV: docs\version 2\feature_documentation\v2_feature_used_unused_audit.csv